

Cálculo Diferencial
Primer Examen Parcial.
04 de Feb. de 2026

1. Determinar el dominio de

$$f(x) = \frac{x+7}{4x^2-x-5}$$

Sol. Igualamos a 0 el denominador

$$4x^2 - x - 5 = 0$$

Factorizamos $4x^2 + (-1)x + (-5)$ Tenemos que encontrar r, s tales que

$$r + s = -1$$

$$rs = 4(-5) = -20$$

$$r = -5, s = 4$$

$$4x^2 - x - 5 = 4x^2 + rx + sx - 5$$

$$= 4x^2 - 5x + 4x - 5$$

$$= x(4x - 5) + (4x - 5)$$

$$= (4x - 5)(x + 1)$$

$$a = 4x - 5$$

$$x + a = a(x + 1)$$

$$4x - 5 = 0$$

$$x + 1 = 0$$

$$x = \frac{5}{4}$$

$$x = -1$$

Conclusión: El dominio es $\mathbb{R} - \left\{\frac{5}{4}, -1\right\}$ ✓

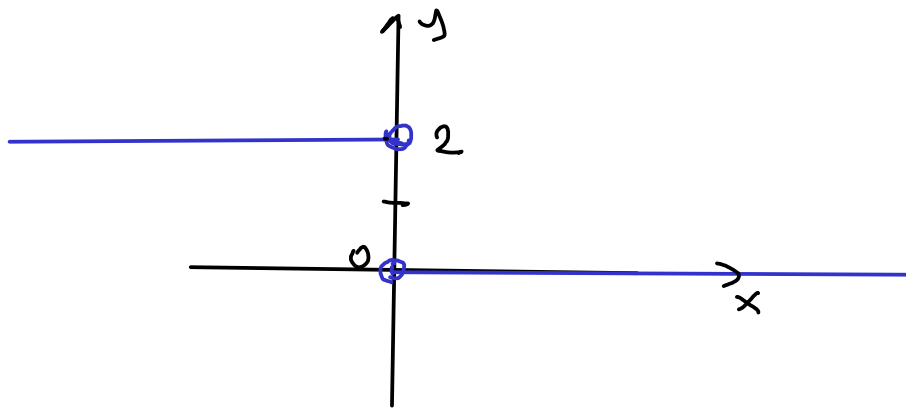
2. Graficar la función

$$f(x) = 1 - \frac{x}{|x|}, \quad x \neq 0$$

Solución.

$$f(x) = \begin{cases} 1 - \frac{x}{x} & , x > 0 \\ 1 - \frac{x}{(-x)} & , x < 0 \end{cases}$$

$$= \begin{cases} 1 - 1 & x > 0 \\ 1 + 1 & x < 0 \end{cases} = \begin{cases} 0 & x > 0 \\ 2 & x < 0 \end{cases}$$



3. Encuentra un valor de a tal que $f(a) = 2$

si

$$f(x) = \frac{x-5}{x+2}$$

Sol.

$$\frac{a-5}{a+2} = 2$$

Resolvemos la ec.

$$a - 5 = 2(a + 2)$$

$$a - 5 = 2a + 4$$

$$a - 2a = 4 + 5$$

$$-a = 9$$

$$a = -9$$

4. calculator got done

$$f(x) = \frac{2x+3}{x-1}, \quad g(x) = \frac{x+3}{x-2}$$

Solution.

$$g \circ f(x) = g(f(x)) = \frac{f(x)+3}{f(x)-2}$$

$$= \frac{\frac{2x+3}{x-1} + 3}{\frac{2x+3}{x-1} - 2}$$

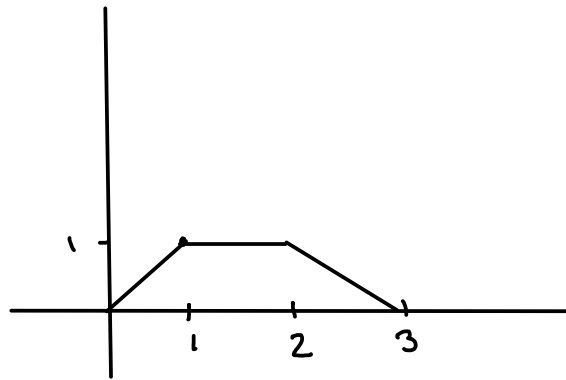
$$= \frac{\frac{2x+3+3(x-1)}{\cancel{x-1}}}{\frac{2x+3-2(x-1)}{\cancel{x-1}}}$$

$$= \frac{2\cancel{x}+3+3\cancel{x}-3}{2\cancel{x}+3-2\cancel{x}+2}$$

$$= \frac{5x}{5} = x$$

$$\frac{\frac{a}{b}}{\frac{c}{b}} = \frac{a}{c}$$

5.



Solución. Encontramos las ecuaciones de los rectos que pasan por $(0,0)$, $(1,1)$, $(2,1)$, $(3,0)$

- De $(0,0)$ a $(1,1)$

$$m = \frac{1-0}{1-0} = 1$$

$$y-0 = m(x-0) = 1 \cdot x = x$$

$$y = x$$

- De $(1,1)$ a $(2,1)$ es constante

$$m = \frac{1-1}{2-1} = \frac{0}{1} = 0$$

$$y-1 = 0(x-1)$$

$$y-1 = 0$$

$$y = 1$$

- De $(2,1)$ a $(3,0)$

$$m = \frac{0-1}{3-2} = -1$$

$$y-1 = m(x-2)$$

$$y-1 = (-1)(x-2)$$

$$y-1 = -x+2$$

$$y = 3-x$$

$$f(x) = \begin{cases} x & x \in [0, 1] \\ 1 & x \in [1, 2] \\ 3 - x & x \in [2, 3] \end{cases}$$